Development and Evaluation of a Deep Learning Based Garment Analysis and Outfit Visualisation System

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ABSTRACT

Personal wardrobe photographs are difficult to use directly for wardrobe management because a single image may contain several garments, uncertain attributes and background clutter. This study develops and evaluates the computer-vision and visualisation subsystem of a Smart Wardrobe mobile web application. The subsystem converts a validated wardrobe photograph into separate, reviewable garment candidates; stores only user-confirmed records; and produces an AI-generated appearance preview for supported outfits. An iterative development approach was used to integrate instance segmentation, optional virtual try-off reconstruction, visible attribute suggestions, visual embeddings, protected media storage and virtual try-on behind model-independent interfaces. The workflow deliberately retains observed segmentation crops and requires user review because generated images and visual predictions can be incorrect. Held-out DeepFashion2 evaluation produced box mAP50 of 0.753 and mask mAP50 of 0.732. On a fixed 30-pair DressCode audit, virtual try-off reconstruction achieved mean SSIM of 0.855, LPIPS of 0.171 and DISTS of 0.188; whole-outfit virtual try-on achieved 0.912, 0.072 and 0.076 respectively. These metrics measure paired image similarity, not physical fit. The resulting subsystem provides a traceable route from image upload to confirmed wardrobe records and an explicitly labelled visual preview, while preserving user control where model output is uncertain.

Keywords: digital wardrobe, garment segmentation, virtual try-off, virtual try-on, computer vision

1. INTRODUCTION

Personal wardrobes are commonly stored as unstructured photographs or scattered physical items. Consequently, a user may not know which garments are available, how an image should be represented as separate records, or whether an AI-generated garment image faithfully preserves the original item. Digital wardrobe systems require a workflow that can extract usable data from ordinary photographs without presenting probabilistic predictions as facts.

DeepFashion2 demonstrates the importance of instance-level clothing detection and segmentation in unconstrained images (Ge et al., 2019), while Fashionpedia provides a vocabulary for visible garment attributes (Jia et al., 2020). Virtual try-on research shows that diffusion-based synthesis can create visually plausible results, but it also introduces risks around occlusion, alignment and altered garment details (Han et al., 2018; Lee et al., 2022). The research gap addressed here is therefore not simply garment recognition or image generation in isolation, but their controlled integration into a private wardrobe workflow with traceability and user review.

This study focuses on three objectives: to implement a multi-garment processing workflow that creates separate candidate records from one wardrobe image; to create editable, evidence-linked garment representations for storage and downstream use; and to generate a whole-outfit visual preview only from user-owned, confirmed garments and an authorised personal photograph. The contribution is a working, model-aware subsystem with explicit failure states, protected media, reversible review decisions and bounded evaluation evidence.

2. METHODOLOGY

2.1 Iterative subsystem development

An iterative and incremental approach was selected because the software workflow and AI components required repeated refinement. The implementation uses a mobile web front end, a FastAPI application layer, protected media storage and PostgreSQL-compatible persistence. Model adapters isolate segmentation, virtual try-off, attribute extraction and virtual try-on from the application services, allowing the workflow contract to remain stable while model behaviour is evaluated.

2.2 Controlled garment processing workflow

A user uploads a validated JPEG or PNG photograph. The system normalises orientation, validates decoded dimensions and creates a persistent processing job. Instance segmentation produces zero or more candidates, each with a category, confidence, source-space mask, bounding box and segmentation-guided crop. The crop remains retained evidence. Optional TryOffDiff reconstruction creates a separate product-style candidate; it never replaces the observed crop automatically. A review page displays each candidate individually, pre-fills supported visible attributes, allows correction through controlled selections and lets the user accept or reject the candidate and choose the preferred image. Accepted candidates are saved atomically as garments or as the members of one outfit.

2.3 Representation and outfit visualisation

The subsystem stores reviewed category, colour-related suggestions, pattern, visible silhouette, optional neckline and sleeve labels where applicable, and a 768-dimensional visual embedding. Fields that cannot be established reliably from an image remain editable. The Fashionpedia-derived predictions are therefore suggestions rather than claims about material, physical fit or hidden construction. For visualisation, the system validates the selected outfit and person image, snapshots the reviewed garment media in a queued request, and applies the supported garments sequentially through a pinned CatVTON runtime. This enables upper, lower and dress layers to be processed as an outfit while preserving order and provenance. Outputs are labelled as AI-generated appearance previews, not size or fit predictions.

2.4 Evaluation approach

Evaluation combined functional acceptance tests, held-out segmentation metrics, blind attribute checks and paired image reconstruction metrics. The segmentation evaluation used the DeepFashion2 validation conversion audit. The reconstruction audits used a fixed-seed, stratified 30-pair sample from the officially authorised DressCode test manifests, with ten upper-body, ten lower-body and ten dress examples. SSIM, LPIPS and DISTS were reported together with qualitative failure labels. These reconstruction metrics quantify similarity to paired reference images; they do not measure garment size, comfort, physical drape or real-world fit.

3. RESULTS

The implemented workflow supports upload progress, cancellation, per-candidate review and transactional persistence. The audit labels below refer to the report's functional-requirement (FR) and AI-requirement (AIR) identifiers. They are acceptance statements for system behaviour and evidence, rather than separate measures of model accuracy. The requirement scope is stated explicitly before the numerical model results.

Audit	Requirement scope	What passed
O1: 25/25	FR-01-05; FR-10-24; FR-37-39; AIR-01-02	Image entry and validation; instance segmentation; candidate review; atomic garment/outfit saving; owner-scoped wardrobe operations; editing and safe archive; failure/cancellation integrity; dataset conversion and held-out segmentation evidence.
O2: 12 groups	FR-06-09; grouped FR-10-18, FR-20-21 and FR-38; AIR-03-05, AIR-07	Optional VTOFF with crop fallback; structured attributes and 768-D embeddings; review/persistence/ownership boundaries; paired VTOFF metrics; explicit generated-media review; attribute and embedding failure evidence; reproducible experiment records.
O3: 18/18	FR-18; FR-25-39; AIR-06-07	Visualisation entry points; source and person-image validation; protected output; visible lifecycle states; AI and physical-fit notice; retry, fail-closed completion, cancellation and provenance; VTON runtime evidence.

Table 1. Requirement-audit scope and accepted behaviour.

The following table reports the principal measured model and representation results.

Evaluation	Result	Interpretation
Held-out segmentation	Box mAP50 0.753; mask mAP50 0.732	Useful candidate evidence, but review remains necessary.
Visible attributes	Colour temperature accuracy 89.7%; primary colour 35.9%; dominance 38.5%	Weak predictions are retained only as editable suggestions.
VTOFF reconstruction, n = 30	SSIM 0.855; LPIPS 0.171; DISTS 0.188	Generated reconstruction outperformed the segmentation-crop baseline on the paired audit.
VTON reconstruction, n = 30	SSIM 0.912; LPIPS 0.072; DISTS 0.076	Bounded paired image-similarity evidence across upper, lower and dress groups.
Visual embeddings	Top-1 category retrieval 47.1%; macro F1 46.6%; MRR 0.601	Embeddings are available for downstream retrieval, not evidence of recommendation quality.

Table 2. Summary of held-out and paired evaluation evidence.

The VTOFF reconstruction exceeded the retained segmentation-crop baseline by 0.072 mean SSIM and reduced LPIPS by 0.137 in the sampled comparison. However, manual inspection identified altered colours, prints, necklines, sleeves and hems in difficult cases. Consequently, VTOFF is offered only as an optional reviewed alternative. The VTON audit produced stronger similarity metrics on its paired sample, but dresses were the weakest group and visible failures included altered drape, necklines and patterns. The product therefore blocks unsupported outerwear layering and retains a physical-fit disclaimer for every generated visualisation.

4. DISCUSSION

The results support the central design decision: AI output should be useful without becoming authoritative. Segmentation supplies the evidence required to construct reviewable candidate records, while the user remains responsible for accepting a garment, correcting descriptive fields and choosing between the observed crop and an optional generated reconstruction. This approach is particularly important because the lower accuracy of primary-colour and dominance predictions shows that a convenient automated field can still be unsuitable for automatic storage.

The evaluation also establishes clear boundaries. The 30-pair reconstruction samples are development-size audits, not full benchmarks or user studies. The image metrics do not demonstrate physical fit, sizing accuracy, comfort or reliable recovery of hidden details. The current VTON workflow supports whole-outfit processing by sequential application, but it does not yet have paired ground truth for multi-garment outfits. Future work should extend multi-garment evaluation, improve difficult garment categories and conduct usability testing with representative wardrobe owners.

5. CONCLUSION

This study developed and evaluated the computer-vision, garment-representation and outfit-visualisation subsystem of a Smart Wardrobe application. The subsystem transforms wardrobe photographs into separate, reviewable candidates; persists only user-confirmed garments and outfits; and generates traceable, explicitly labelled visual previews from authorised media. The evidence shows that the workflow operates as designed within its stated scope. Its contribution is a controlled, user-reviewable integration of deep-learning outputs into a personal wardrobe system, not a claim of perfect fashion understanding or physical-fit prediction.

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